Model Development Phase Template

Date	4 April 2026
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Project Title	Restaurant Recommendation System
Maximum Marks	10 Marks

Initial Model Training Code, Model Validation and Evaluation Report Initial

Model Training Code (5 marks):

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.feature_extraction.text import TfidfVectorizer
df_percent.set_index('name', inplace=True)
indices = pd.Series(df_percent.index)

# Creating tf-idf matrix
tfidf = TfidfVectorizer(analyzer='word', ngram_range=(1, 2), min_df=0, stc
tfidf_matrix = tfidf.fit_transform(df_percent['reviews_list'])

cosine_similarities = linear_kernel(tfidf_matrix, tfidf_matrix)
```

Model Validation and Evaluation Report (5 marks):

Model	Summary	Training and Validation Performance Metrics

Model 1	Content-based Recommendation	**Training Metrics -**None (unsupervised, no explicit training phase) **Validation Metrics -** None (recommendations are inspected manually)